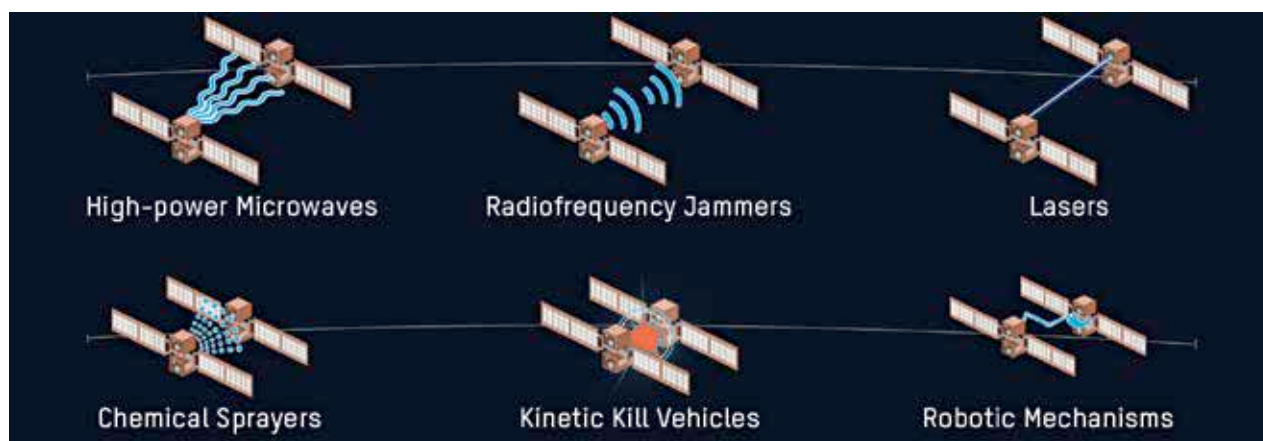




A summary of space threats

Credit: DIA/D3 Design

Emerging orbital threats include satellites that interact with other satellites in space, with jammers, lasers targeting the sensors to make them useless, satellites that spray chemicals to degrade other satellites, and even kinetic impactors that just ram themselves kamikaze style into other satellites. Deorbital satellites with electromagnetic tethers or repair satellites with robotic arms, all have the potential of being weaponized in the space environment, being able to physically pull targets out of orbit, or even tear them apart piece by piece. There is a danger that countries may surreptitiously launch satellites for the apparently peaceful purposes of debris removal or servicing other satellites, but can be put to nefarious use if and when the situation demands it. One of the most straightforward ways of countering the technologies that can remove satellites from orbit, is to put more satellites in orbit, but there are other compelling reasons to now use swarms of small satellites in low orbits instead of the big satellites in high geostationary orbits that the militaries tend to use today. .

The march of miniaturisation has led to atomic scale components, where there are now physical limits to how much more circuitry can be packed into so much space. Neural networks need to be trained on a lot of data, theoretically infinite data to be perfect, which is a challenge when it comes to military grade AI because of the lack of availability of datasets. For example an AI algorithm that can recognise any weapon in infrared

imagery would need to be trained on a vast bank of images with each of the weapons tagged by humans, or preprocessors. For an exhaustive list of all weapons in the world, the database would be quite large and impractical. In the space scenario, this is compounded by the size of computers that can be deployed into orbit, as well as heat management, there is very little extra heat that can dissipate in orbit. To train orbital satellites for military applications, a mesh network is needed, a constellation of satellites that can all talk to each other. The US Space Force is in the process of deploying such a network in low Earth orbit, with a planned constellation of hundreds of satellites. The capabilities needed for this constellation is being verified with DARPA's Blackjack program, with a number of technology demonstration satellites that are all in the process of being launched as rideshares along with primary payloads. The Mandrake 1 is a CubeSat loaded with supercomputing chips, the Mandrake 2 mission is a pair of satellites with high bandwidth inter satellite optical links, and Wildcard is a radio payload that will test communications with conventional radio systems from LEO. Then there is the POET, a hosted payload for auto-

nous mission management, to test the ability to fuse mission data on orbit. In August 2021, the Space Force deployed the PIRPL satellite, which uses a multispectral sensor to detect ballistic missiles. All of these together form an optically meshed computer network in orbit. This constellation is meant to provide early warning systems to counter the ASAT capabilities of China and Russia, making it difficult to reduce US capabilities



A modified Prithvi mission used for India's ASAT test in March 2019

Credit: Press Information Bureau

by taking down single satellites. The Blackjack program is the most ambitious undertaking by any country to counter the threat of ASAT weapons, and is essentially a drone swarm in space that provides intelligence to all the drone swarms potentially working in tandem on Earth. **d**